Day 3 Deliverable

Project Title

CodeMentor AI — Multi-Agent Mini Coding Copilot

Brief Abstract

CodeMentor AI is a multi-agent software engineering assistant designed to act like a lightweight coding copilot. The system will use the OpenAI Agents Framework and four specialized agents that collaborate to help users write, review, debug, and test code. A Coordinator Agent will manage the workflow, a Code Writer Agent will generate or modify code, a Code Review & Debugging Agent will analyze code quality and identify issues, and a Testing Agent will generate test cases. The final output will combine the agents' results into a structured response for the user.

This still fits the Day 3 requirement to finalize the title, abstract, goals, expected features, and agent roles.

Key Goals

- Build a four-agent system using the OpenAI Agents Framework.
- Help users generate and modify source code.
- Review code quality and identify possible bugs.
- Suggest fixes and improvements.
- Generate relevant test cases and edge cases.
- Demonstrate collaboration between specialized agents.

Expected Features

- Accept coding questions or programming tasks.
- Generate code from user requirements.
- Modify existing code based on requested changes.
- Review code quality and readability.
- Detect possible syntax and logic issues.
- Suggest corrections and improvements.
- Generate normal and edge-case tests.
- Return one structured final response.

Agent Roles

1. Coordinator Agent

Receives the user request, understands the task, routes work to the appropriate agents, and combines the results into the final response.

2. Code Writer Agent
Generates new code or modifies existing code based on the user's requirements.

Main responsibilities:

- Write functions, classes, or small programs.
- Implement requested features.
- Modify existing code.
- Follow the requested programming language and constraints.

3. Code Review & Debugging Agent
Reviews generated or user-provided code and identifies quality issues or possible bugs.

Main responsibilities:

- Check readability and structure.
- Identify syntax or logic problems.
- Suggest improvements and fixes.
- Explain issues clearly.

4. Testing Agent
Creates tests to verify that the code behaves as expected.

Main responsibilities:

- Generate normal test cases.
- Generate edge cases.
- Provide expected results where appropriate.
- Identify scenarios that should be tested.

Basic Workflow

User Request → Coordinator Agent → Code Writer Agent → Code Review & Debugging Agent → Testing Agent → Coordinator Agent → Final Code + Review + Tests

For requests where the user already has code, the Coordinator can skip the Code Writer and send the code directly to the Review/Debugging and Testing agents.

Basic Guardrails

- Do not claim code was executed unless it was actually executed.
- Clearly distinguish confirmed problems from possible issues.
- Do not expose API keys, passwords, or other sensitive information.
- Generated code should follow the user's stated requirements.
- Each agent should remain focused on its assigned responsibility.